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Integrated Two-Way Coupling of Reservoir Simulation & Geomechanical Models in CCS Operations

Ashok Shinde, Anirban Banerjee, Tim Salter, Satya Perumalla, Irfan Sami, and Jorge Vladimir Fandino, Baker Hughes; Pouria Mousavi, CMG

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Abstract

In a modeled CCS prospect, saline aquifers are assessed for the feasibility of safe CO₂ storage. Two-way coupling between Dynamic Reservoir Simulation and Geomechanical modeling is carried out to understand the interaction between reservoir properties (porosity, permeability, etc.) and geomechanical parameters (stress & strain) due to the pressure changes occurring during CO₂ injection. This paper discusses the detailed workflow, data requirements and detailed analysis for two-way coupled models in.

3D dual grid system is adopted for two-way coupling between reservoir and geomechanical models. Reservoir grids are very fine and represent the area of CO₂ injection, whereas Geomechanical grids are coarser and cover ground surface, overburden, caprock, reservoir, side burdens, and under burden. It is an iterative coupled model where pressure from reservoir simulator is passed on to geomechanical module to compute stress/strain changes, which in turn will update porosity, permeability and compressibility for next time step in the reservoir simulator where fluid flow calculations involving pressure and saturation changes are performed.

Two-way coupled simulations are run for 25-years of CO₂ injection followed by 50-years of observation without injection/production. Results from two-way coupled models are analyzed for CO₂ plume modeling and geomechanical risk assessment. Ground surface deformations and reservoir deformations are also predicted. Reservoir properties changes due to CO₂ injection are also modeled and updated at every time-step. Geomechanical risk assessment e.g. Fault stability and caprock integrity analysis can also be performed using the results of two-way coupled models.

Two-way coupled reservoir & geomechanical models helped us to model interaction between reservoir and geomechanical parameters, thereby accurately predicting reservoir storage parameters and geomechanical risk assessment. Two-way Coupling workflow developed in this study can be applicable to other CCS projects for integrated risk assessment.

Introduction & Workflow

Figure 1(a) represents general workflow for 3D/4D Geomechanical Model. It starts with building structural framework, integrating overburden, reservoir and underburden horizons and faults in a water-tight structural

model. Completed structural model will then be discretized into simple geometric shapes called elements. 3D Geomechanical model is then coupled (one-way or two-way coupling) with varying pressure and/or temperature at different times as per dynamic reservoir simulation model. 3D Finite Element model will predict stress and strain changes through time due to changes in pressure and temperature. These stress and strain changes will be used to evaluate reservoir and surface deformations, fault stability, caprock integrity and associated risk assessment.

Workflow of two-way coupling between Geomechanics & Reservoir Simulation is presented in Figure 1(b). In two-way coupling, reservoir pressure and/or temperature are sent from reservoir simulator to geomechanical module. Stress and strain computed from geomechanical module are used to determine new pore compressibility and absolute permeability. These changes in pore compressibility and absolute permeability are provided as an input for new time-step in reservoir simulator and continues on, for multiple cycles depicting the changes in reservoir pressures.

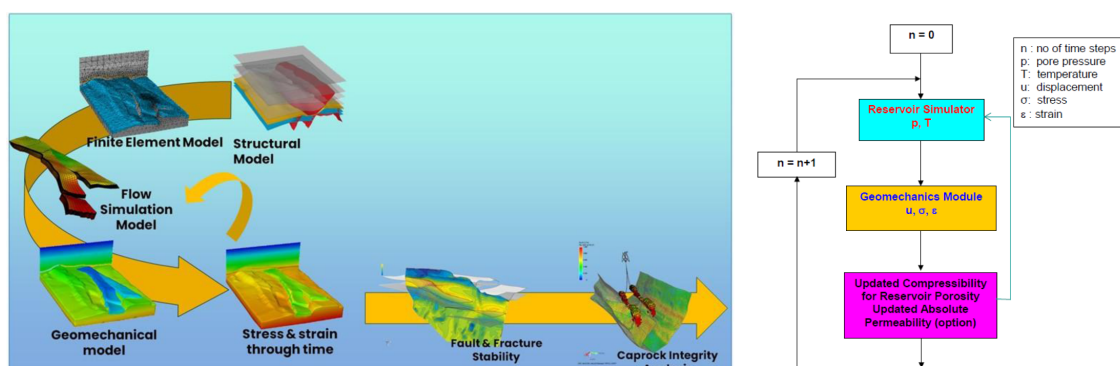


Figure 1—(a) General Workflow for 3D/4D Geomechanical Model (b) Two-way Coupled Modeling Workflow

Dual Grid System

For two-way coupling, dual grid system is implemented where Geomechanical grid and Reservoir simulation grid are independent as shown in Figure 2. Geomechanical grids are extended to overburden, under-burden & side-burdens. Since Geomechanical grids have extended area, this can lead to a greater number of grid cells and hence, Geomechanics grids with optimum grid resolution can help in two-way coupling. This in turn can help in optimization of computation run time. Both these independent grid results can also be plotted together. Results from single and dual-grid systems have been found to be very comparable.

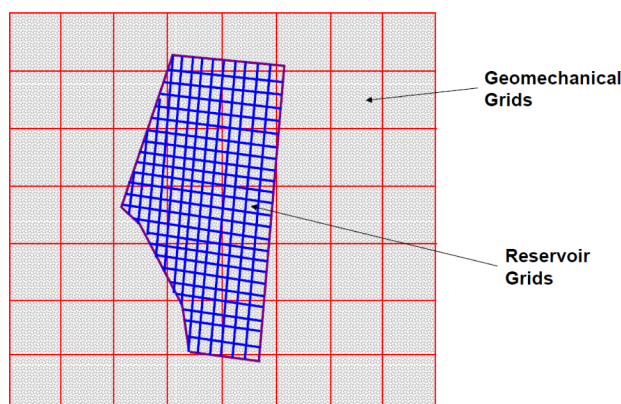


Figure 2—Dual Grid system for Two-way Coupling: Fine-scale and localized reservoir simulation grid & Coarse-scale and extended Geomechanical grid.

Porosity Changes in Two-way Coupling

Geomechanical model computes stress and strain changes due to reservoir pressure and/or temperature changes. Volumetric strain increases due to CO₂ injection, due to expansion of rock mass. These volumetric strain changes can be related to changes in porosity.

Following are the key assumptions made at start of the 2-way coupling exercise:

- Bulk volume of reservoir grid is constant.
- Reservoir pore volume = True pore volume.
- Reservoir porosity is used in fluid flow computation instead of true porosity.

$$\begin{aligned} \text{Reservoir porosity } \phi^* &= \frac{\text{Current pore volume } (V_p)}{\text{Initial bulk volume } (V_b^0)} \\ \text{True porosity } \phi &= \frac{\text{Current pore volume } (V_p)}{\text{Current bulk volume } (V_b)} \end{aligned} \quad \longrightarrow \quad \frac{\phi^*}{\phi} = \frac{V_b}{V_b^0}$$

The conservation of fluid in a deformable porous media can be expressed by:

$$\partial / \partial t [\phi \rho_f (1 - \varepsilon_v)] - \nabla \cdot (\rho_f k / \mu \cdot [\nabla p - p_f b]) = Q_f$$

where, ϕ = "True porosity" and ϕ^* = "Reservoir porosity"

Reservoir porosity is related with true porosity as a function of volumetric strain changes due to changes in reservoir pressure and temperature.

Permeability Changes in Two-way Coupling

Permeability is related as a function of volumetric strain changes due to changes in reservoir pressure and temperature. Permeability changes are calculated using the below empirical equation (Li & Chalaturnyk):

$$\frac{k}{k_0} = \exp(C_{n1} \varepsilon_v)$$

Where,

K_0 : Initial permeability

C_{n1} : Parameter for each rock type

ε_v : Volumetric strain

3D Geomechanical Grid – Structural Framework

3D Geomechanical structural model is a watertight model, which is a representation of sub-surface from ground level by integrating all formation horizons as well as faults. In this study, 3D Geomechanical grid is built by integrating overburden, reservoir, under burden and side burdens as shown in [Figure 3\(a\)](#). 3D Reservoir grids are presented in [Figure 3\(b\)](#) which is limited only to reservoir area where CO₂ injection is planned.

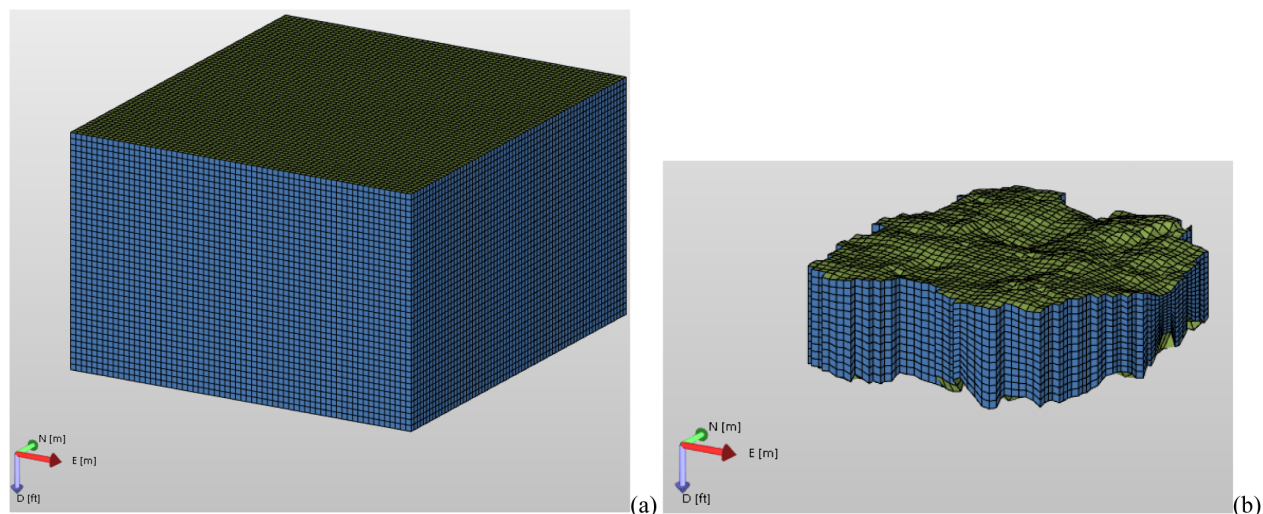


Figure 3—(a) 3D Geomechanical Grid including over burden, reservoir, under burden and side burdens (b) 3D Reservoir Grid

3D Geomechanical Rock Properties

Young's Modulus & Poisson's ratio are elastic properties of the rocks, this model uses Linear Elastic model. Young's modulus and Poisson's Ratio varies as a function of depth representing mechanical layering. Figure 4(a) and Figure 4(b) shows Young's Modulus & Poisson's Ratio for the Geomechanical model.

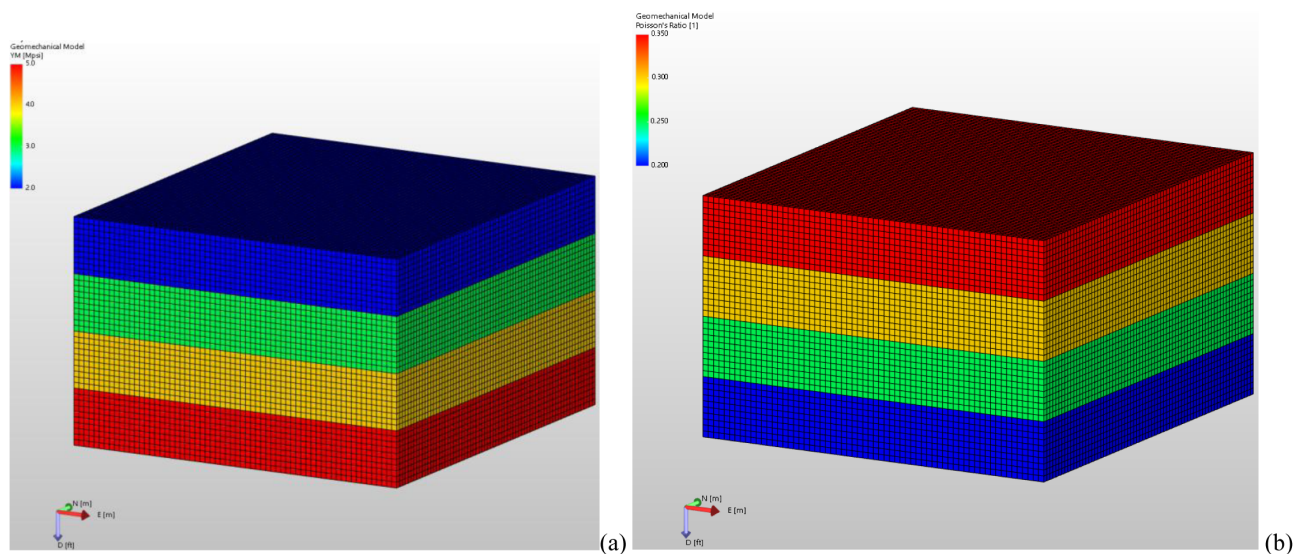


Figure 4—(a) Young's Modulus and (b) Poisson's Ratio cube in 3D volume representing from ground level to below the reservoir layers.

3D Pore Pressure & In-situ Stresses

Initial reservoir pressure is assumed to be hydrostatic. Pore pressure gradient of 0.43 psi/ft is assumed, and pore pressure cube is presented as shown in Figure 5(a). Vertical stress is integration of density log from surface to reservoir TD. Vertical stress cube is built using assumed stress gradient of 1 psi/ft as shown in Figure 5(b).

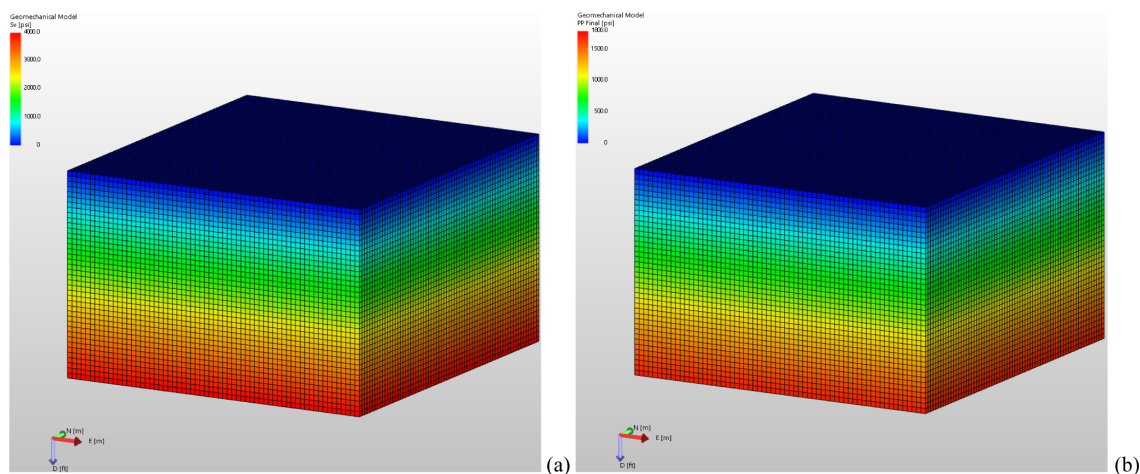


Figure 5—(a) Pore Pressure Cube and (b) Overburden Stress cubes in 3D volume representing from ground level to below the reservoir layers.

Results of Two-way Coupling

Two-way coupled Geomechanical & Reservoir engineering models are simulated for the following case. One CO₂ injector well is considered in the simulation, injecting for 25-years with 45-years of observance. Cumulative gas mass (kg) and gas injection rates (m³/day) are presented in Figure 6.

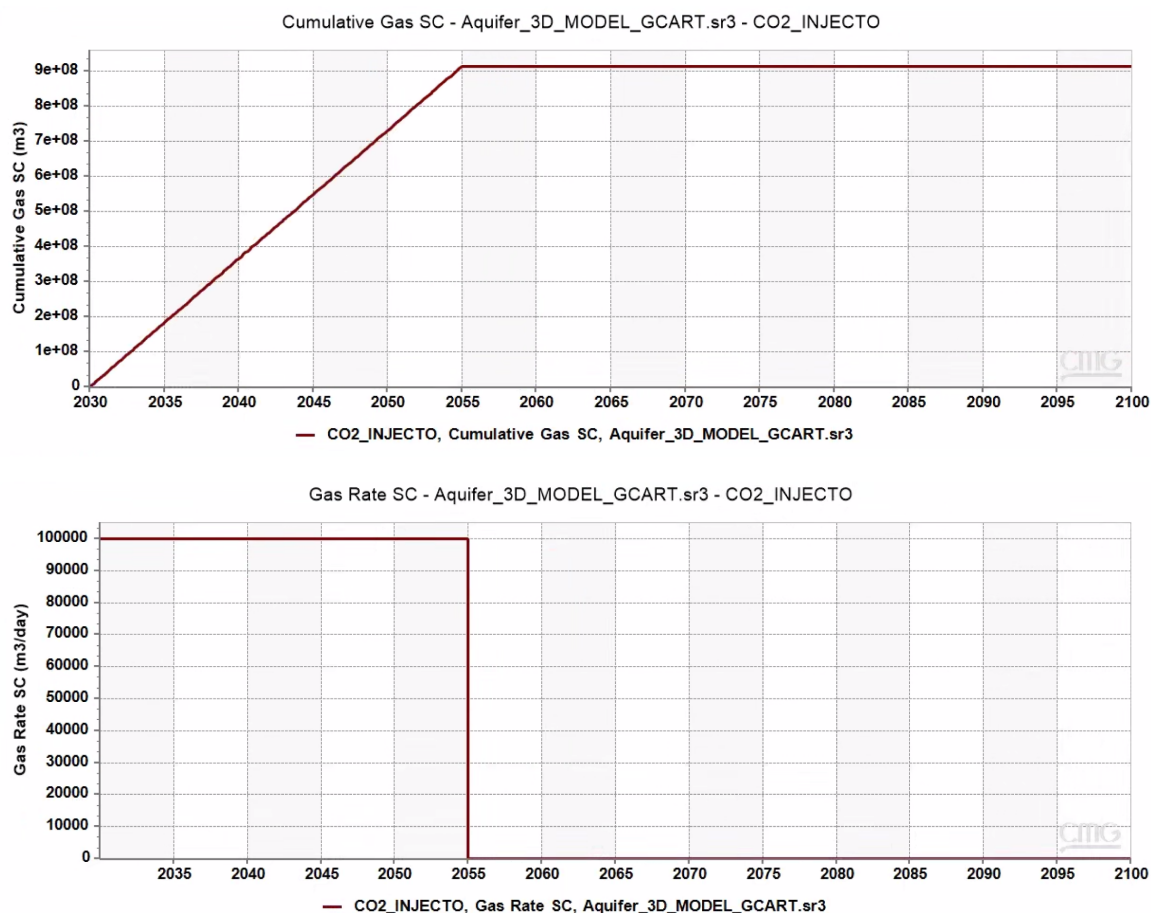


Figure 6—(a) Cumulative Gas Mass (b) Gas Injection rate for the study reservoir

Two-way coupled model not only predicts stress/strain changes due to reservoir pressure changes but also computes porosity/permeability changes and updates compressibility to the next time-step. These parameters will be an input to reservoir simulator for further processing. Reservoir pressure changes are modeled and presented as shown in Figure 7.

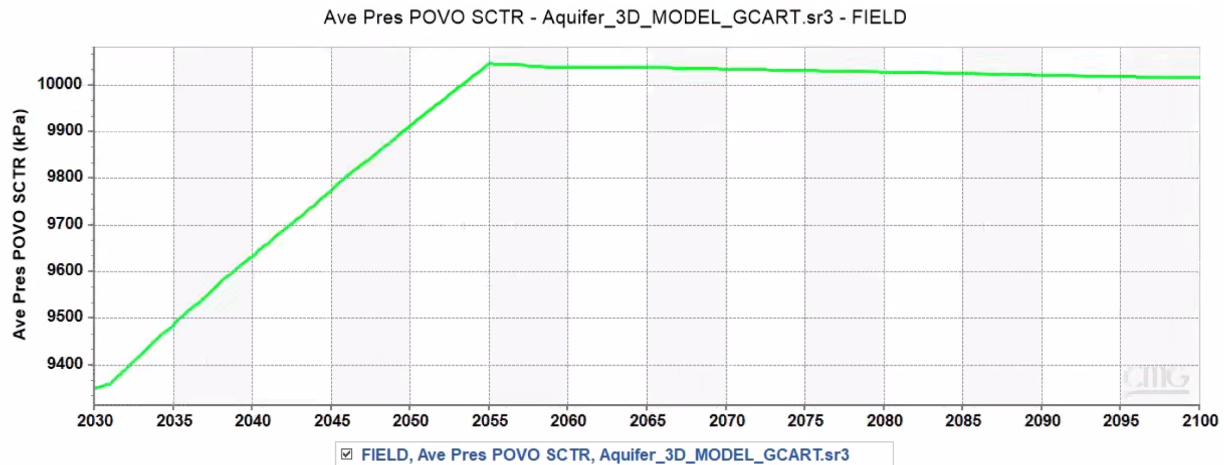


Figure 7—Reservoir pressure changes in the study reservoir (design base case)

Reservoir & Ground Surface Deformations

CO₂ injection leads to increase in pressure in the reservoir, which leads to volume expansion of rock and lead to reservoir expansion. Magnitude of expansion depends on amount of pressure changes in reservoir, mechanical properties of reservoir rock, thickness of reservoir, stiffness of overburden, etc. Figure 8 shows ground surface deformations at selected time-steps. Absolute magnitude of ground surface deformations (upheave) due to CO₂ injection increases with time.

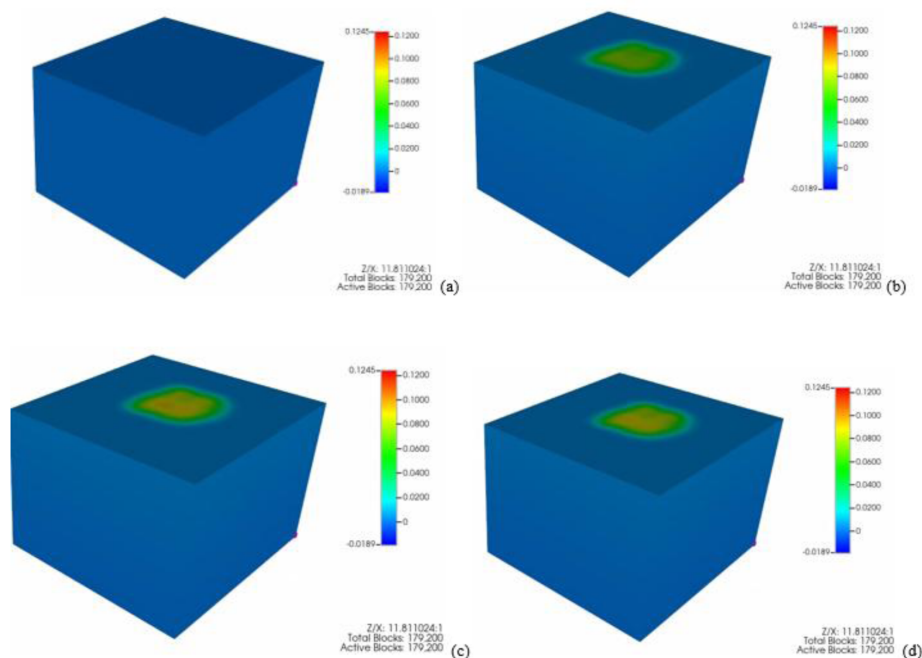


Figure 8—Ground Surface Deformation due to CO₂ injection at selected time intervals

Figure 9 shows cross-section view showing vertical displacement at the top of reservoir and overburden/underburden at selected time-steps. Absolute magnitude of vertical displacements (expansion) at top of reservoir due to CO₂ injection increases with time.

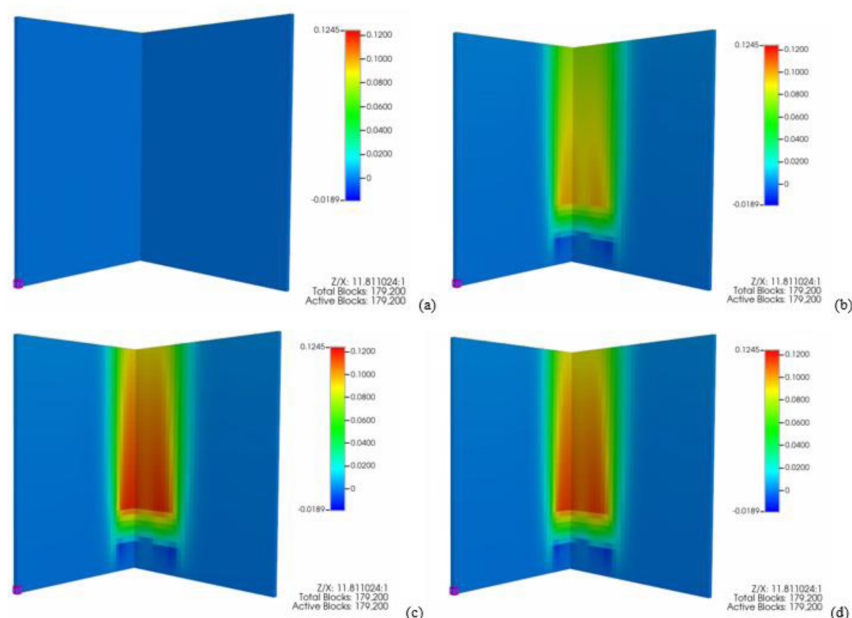


Figure 9—Cross-section showing vertical displacement in reservoir, overburden and under-burden due to CO₂ injection

Volumetric Strain Changes

Figure 10 shows volumetric strain changes in the reservoir due to CO₂ injection at selected time-steps. This volumetric strain changes are related to changes in porosity and permeability as discussed earlier. In this case, increase of porosity and permeability due to CO₂ injection are expected.

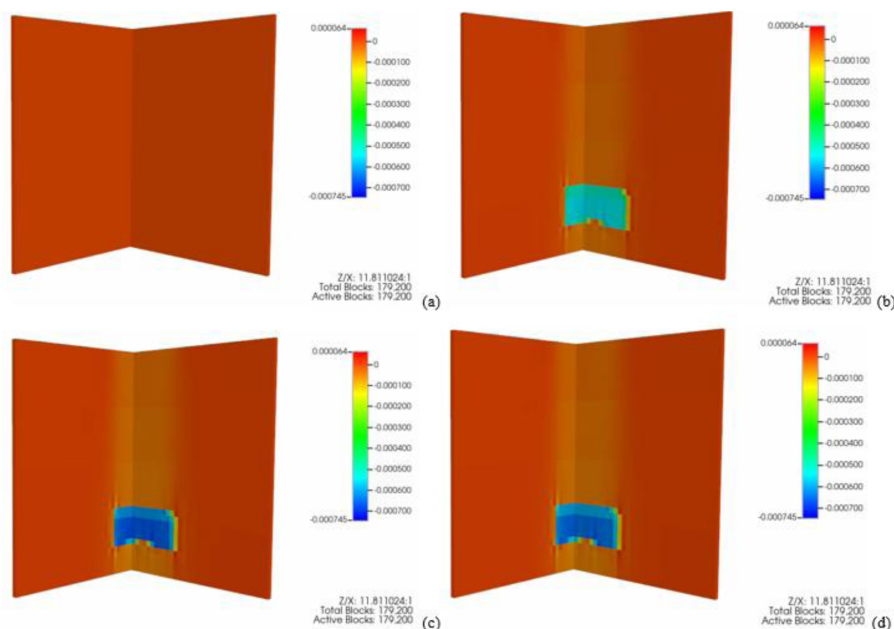


Figure 10—Cross-section showing volumetric strain changes in the reservoir due to CO₂ injection at selected time intervals

Porosity & Permeability Changes

Figure 11(a) & (b) shows porosity & permeability changes near injector well-A (at selected grid blocks). Increase in porosity and permeability are observed due to CO₂ injection. However, permeability improvement in two-way coupling is observed due to CO₂ injection.

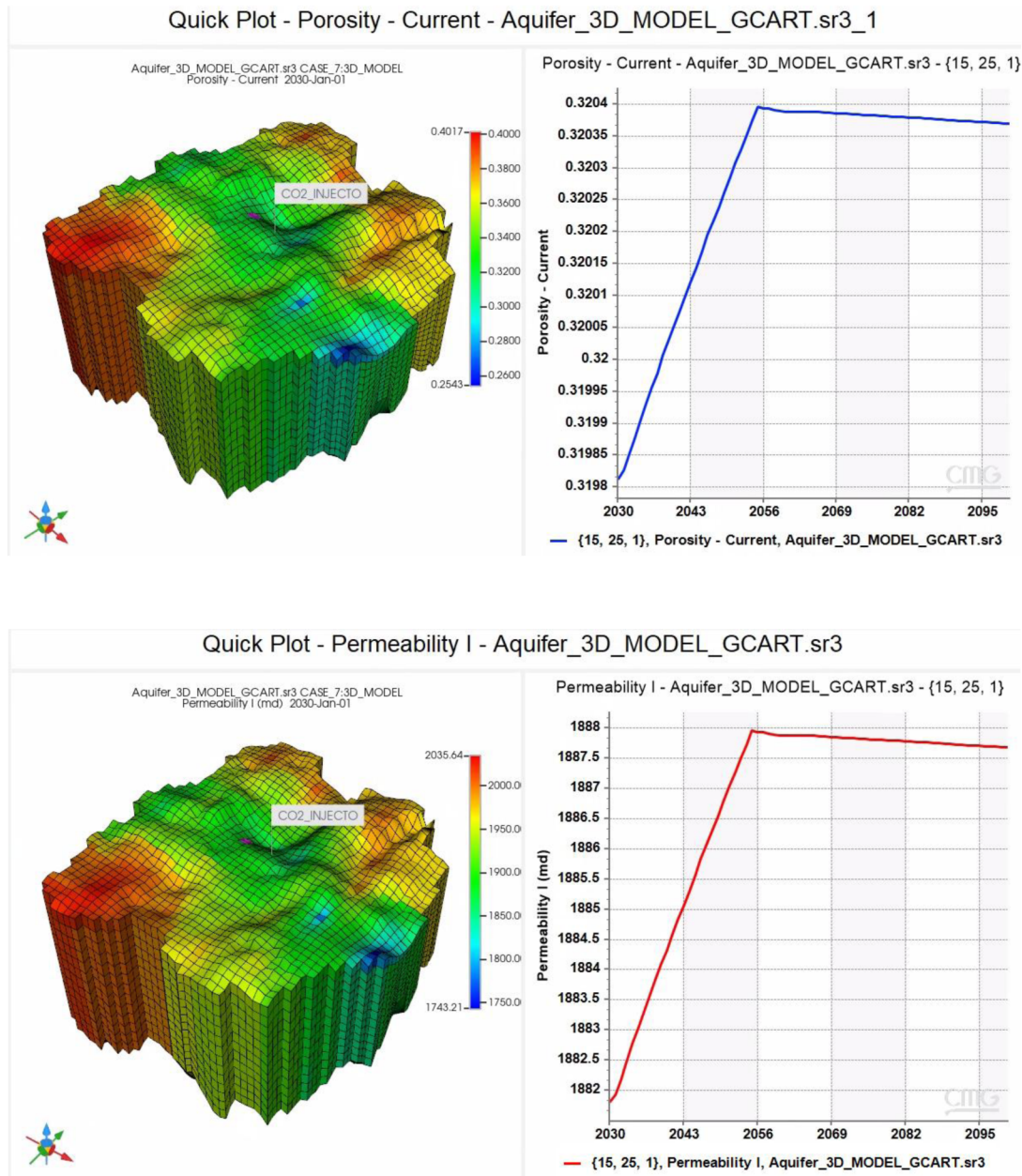


Figure 11—Porosity & Permeability increase due to CO₂ injection in reservoir model, on selected block 15, 25, 1

Stress Path

CO₂ injection leads to an increase in pressure in the reservoir, which leads to volume expansion of rock and the resulting reservoir expansion. In the sub-surface, where the rocks are not allowed to deform freely, it leads to an increase of total stresses due to increasing in reservoir pressure. Changes in stress due to changes in reservoir pressure is also called stress path. It is very important to predict stress changes in the reservoir and caprock as a function of reservoir pressure changes due to CO₂ injection and its impact on fault stability and caprock integrity.

Stress path can also be quantified from finite element modeling which is presented in this study.

$$\text{Stress Path} = \frac{\text{Changes in stresses (Shmin or SHmax)}}{\text{changes in reservoir pressure (PP)}}$$

Figure 12 presents stress path for Shmin and SHmax at 2100 time-step for study reservoirs respectively. Stress path of 0.55-0.65 is predicted. This suggests that for every 1000 psi reservoir injection, stresses are increased by 550-650 psi. Stress path is applied in calculating stress changes due to CO₂ injection in caprock integrity & fault slip analysis.

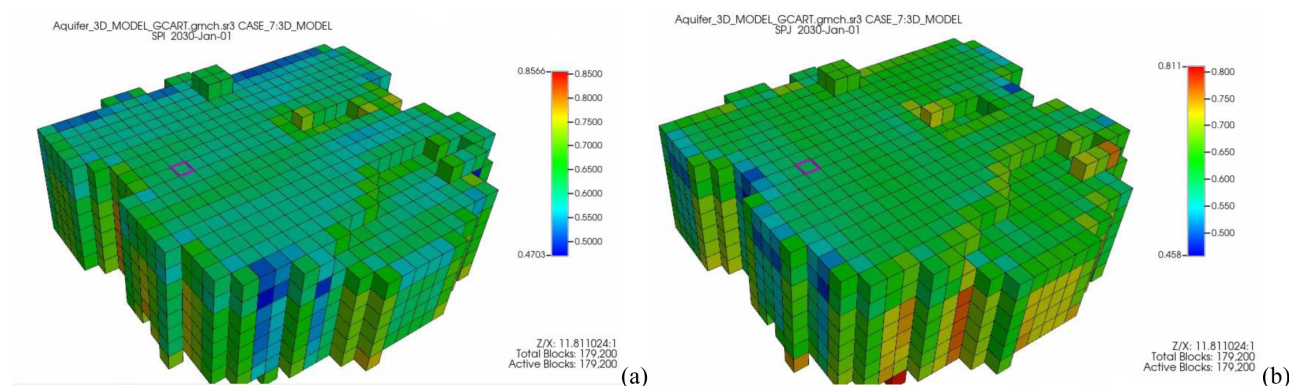


Figure 12—Stress path for Shmin and SHmax at 2100 time-step for the study Reservoir

Summary of Two-Way Coupled Modelling

Two-way coupled simulations are run for 25-years of CO₂ injection followed by 45-years of observation. Results from two-way coupled models are analyzed for CO₂ plume modeling and geomechanical risk assessment. Ground surface deformations and reservoir deformations are also predicted. Reservoir properties changes due to CO₂ injection are also modeled and updated at every time-step. Geomechanical risk assessment e.g. Fault stability and caprock integrity analysis can also be performed using the results of two-way coupled models. Based on the results of two-way coupled model, comprehensive MMV (Measurements Monitoring & Verification) plan can also be prepared for derisking the overall integrity during CCS operations.

Two-way coupled reservoir & geomechanical models helped us to model interaction between reservoir and geomechanical parameters, thereby accurately predicting reservoir storage parameters and geomechanical risk assessment. Two-way Coupling workflow developed in this study can be applicable to other CCS projects for integrated risk assessment.

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